

PARKWISE DATA ARCHITECTURE

Comprehensive Data Acquisition & Engineering Strategy for Nairobi Urban Parking

Project: Parkwise | **Target Area:** Nairobi (CBD, Westlands, Kilimani, Upper Hill) | **Status:** Data Requirements Specification

To deliver an intelligent parking discovery and routing platform, **Parkwise** combines static geospatial boundaries, commercial tariff structures, qualitative sentiment, user interaction telemetry, and simulated real-time operational data. Below is the technical roadmap detailing what data is required, its role within the system, and exact extraction mechanisms.

1. Primary Data Acquisition Matrix

CATEGORY	SPECIFIC DATA REQUIRED	EXACT EXTRACTION METHODOLOGY
SPATIAL & GEOLOCATION	- Spot / Facility Name - Exact GPS Coordinates (Lat/Long) - Type (On-street vs. Off-street / Garage) - Total Capacity / Bay Counts - Operating Hours	Overpass API (OSM): Query amenity=parking over Nairobi bounding box to fetch open GeoJSON data. Google Places / Mapbox Search API: Programmatic geocoding and metadata extraction for commercial lots.
PRICING & TARIFFS	- Rates by vehicle classification - Tariff structure (Flat daily vs. Hourly) - Accepted payment channels (M-Pesa, Cash, USSD) - Penalty / Overstay rules	County Portals: Scrape Nairobi City County published schedules (e.g., NairobiPay *647# zone rates). Private Vendors: Sample and parse tariff rate cards from operators (KAPs, PayTech) at key malls (Yaya Centre, Sarit, Junction).
QUALITATIVE & SENTIMENT	- Raw review text - Numerical star ratings - Timestamps - Feature tags (Security, Accessibility, Attendants, Price)	Google Maps Web Scrapers: Browser automation (Playwright/Selenium) or browser extensions (Instant Data Scraper). Social Media (X/Twitter): Scrape public posts via snsrape filtering keywords like #NairobiTraffic, parking Nairobi.
TRAFFIC & OCCUPANCY	- Surrounding road congestion index - Estimated travel delay / distance - Dynamic spot occupancy rates	TomTom / Google Distance Matrix API: Fetch real-time traffic speeds. Synthetic Poisson / ARIMA Simulation: Model temporal occupancy based on time-of-day, day-of-week, and venue event schedules.
USER PREFERENCES	- Vehicle dimensions / type - Max budget constraint	

CATEGORY	SPECIFIC DATA REQUIRED	EXACT EXTRACTION METHODOLOGY
	• Intended parking duration • User feedback (Upvote/Downvote)	• Application Telemetry: Log directly via Streamlit / UI input form parameters and user interaction events.

2. Data Utilization Pipeline inside Parkwise

Each data source plays a specific functional role within the multi-layered Parkwise application stack:

- **Layer 1 (Discovery & Filtering):** Uses Spatial and Pricing data to return parking options within a 1km radius of the user's destination matching their vehicle type and max hourly budget.
- **Layer 2 (Aspect-Based NLP Scoring):** Analyzes qualitative review text using Natural Language Processing (TF-IDF / Sentiment Analysis) to generate trust and safety scores (e.g., *Security Rating: 4.5/5, Pricing Transparency: 2.1/5*).
- **Layer 3 (Dynamic Recommendation Engine):** Combines simulated real-time occupancy with travel time APIs to recommend the optimal facility based on convenience, cost, and availability likelihood.
- **Layer 4 (Analytics Dashboard):** Consolidates telemetry and reviews into interactive Streamlit/Tableau visualizations showing demand hotspots and operational bottlenecks for urban planning insights.